BHJ and by using non-conventional donor/acceptor combinations.

Commercialisation

Many multinational companies are interested in this newly emerging solar technology. Researchers are putting in a lot of effort in this field to make it better and efficient. Today, efficiency of an organic solar cell is ten per cent, which is way behind silicon solar cells', but it is in progress. A number of organic photovoltaic manufacturing companies have been established and are receiving good feedback from the market. Some of them have launched their products commercially in the market, too.

Advantages

Some advantages are given below:

- Low weight and flexibility of photovoltaic modules
- Semi-transparency
- Easy integration into other products
- New market opportunities, for example, wearable photovoltaics
- Significantly lower manufacturing costs compared to conventional inorganic technologies
- Manufacturing of organic photovoltaics is a continuous process using state-of-the-art printing tools
- Short energy payback times and low environmental impact during manufacturing and operations

Most of the advantages listed above also apply to solar cells based on vapour-deposited small-molecule absorbers. This suggests that organic photovoltaics have the potential to be a disruptive technology within the photovoltaic market. The bright outlook has initiated many research and development activities, and substantial progress has been made in increasing the PCE of solutions-processed organic photovoltaic in the last few years.

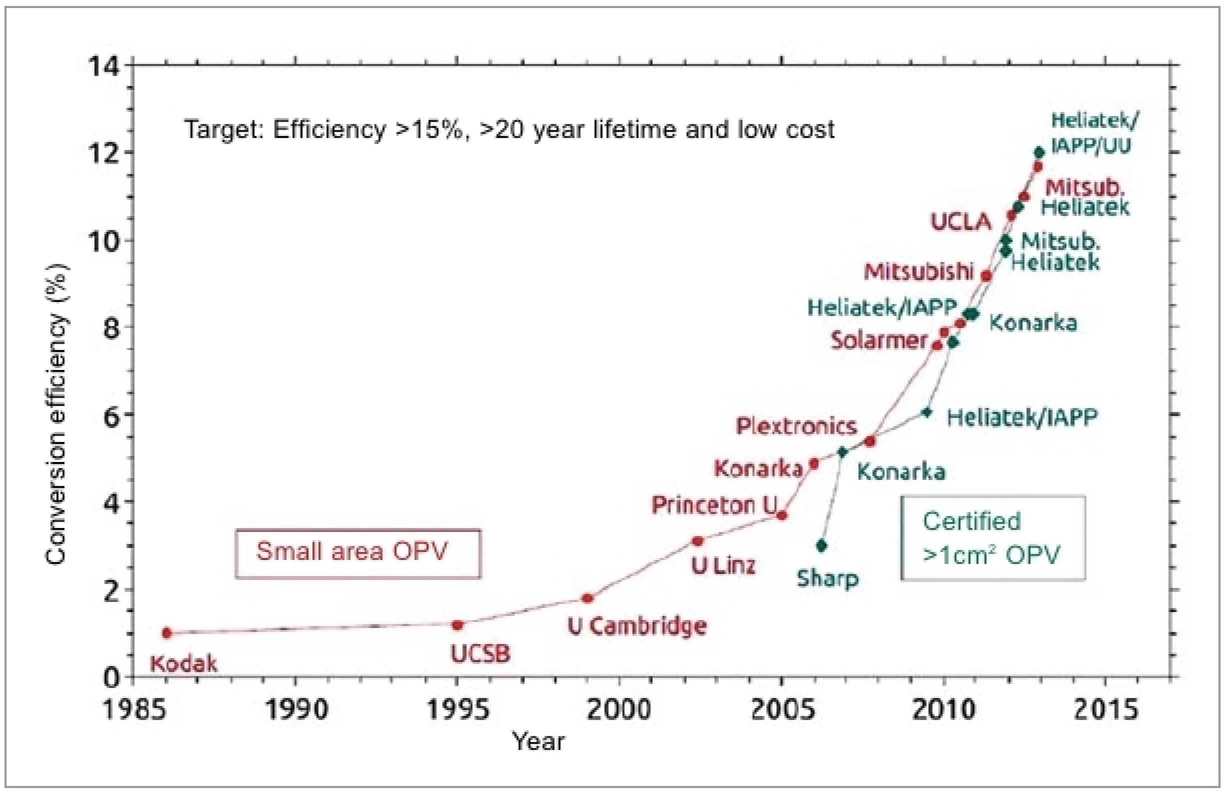

Fig. 8: Power conversion efficiency scenario

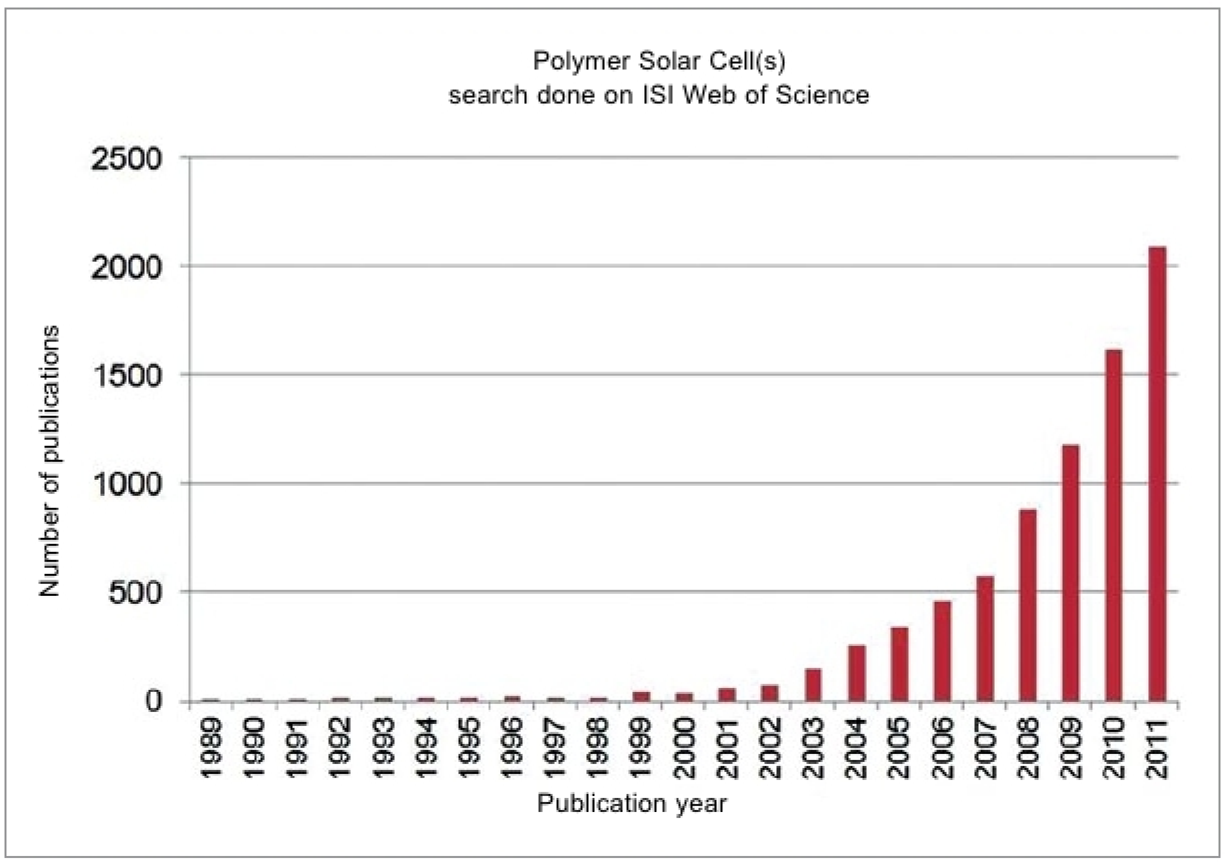

Fig. 9: Research report survey for 22 years

Conclusion

To summarise, bulk heterojunction organic solar cells represent a promising technology that could be an important player in the future photovoltaics market. Substantial research and development efforts are required to bring the technology to better performance levels.

The flexibility offered by organic chemistry to design semiconductors and engineer interfaces to other inorganic or organic materials would offer various opportunities to explore third-generation concepts. This advancement in technology will hopefully prove fruitful for a green earth and, subsequently, reduce greenhouse gases.

Inventions and researches in organic solar cells are noteworthy due to which the predicted future of organic solar cells is extremely bright. Many multinationals are interested in investing in such constructive ideas. Tesla is trying to make rooftops using solar cells, which will be a good source of energy and rigid enough to protect roofs as well.

Also, as mentioned above, organic solar cells are at present semi-transparent, and efforts are being made to make these entirely transparent for installation in home windows. Organic solar cells will then play a vital role in constructing a better and eco-friendly future. EFY